

Trigonometric Functions

Rail Engineering Mathematics

Sheffield Hallam University

Lecture 2

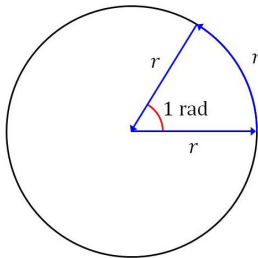
Learning Outcomes

- Introducing trigonometric functions and radian measure.
- Calculate lengths and angles using right-angled triangle rules.
- Rules for calculating lengths and angles in more general triangles.
- General form of trigonometric functions.
- Plot general trigonometric functions using MATLAB.
- Solve equations involving trigonometric functions.

Degrees and Radians

Degrees are a measure of angle, where one full rotation is equivalent to an angle of 360° .

Radians are more commonly used throughout mathematics except when discussing angles of shapes. One radian is the angle between two radii that create a circular arc with length equal to the radius:



Degrees and Radians

Since the circumference of a circle has a length of 2π radii, there must be 2π radians in a full rotation.

Therefore:

Radians - Degrees Conversion

$$2\pi \text{ rad} = 360^\circ$$

and also

$$1 \text{ rad} = 57.3^\circ \text{ to 1 d.p.}$$

Radians	Degrees
0	0
$\frac{\pi}{2}$	90
π	180
$\frac{3\pi}{2}$	270
2π	360
1	57.3

The Trigonometric Functions

Trigonometric functions take an angle (usually measured in radians) as their input.

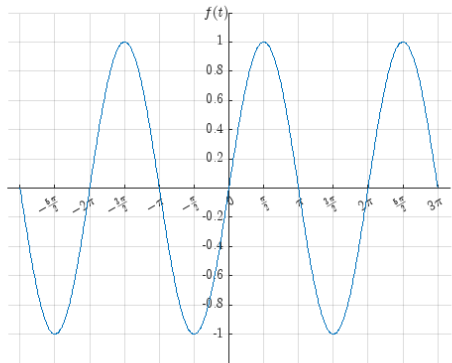
They are **periodic**, meaning that they repeat a pattern indefinitely.

There are three main functions of this kind, and we need to be familiar with their appearance:

- $y = \sin(t)$
- $y = \cos(t)$
- $y = \tan(t)$

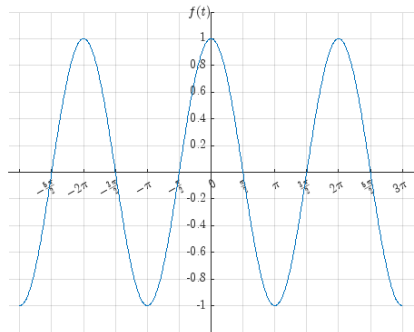
Sketching $\sin(t)$

$\sin(t) = 0$ when $t = 0$. It varies between -1 and 1 and takes 2π rad (or 360°) to complete one full cycle.



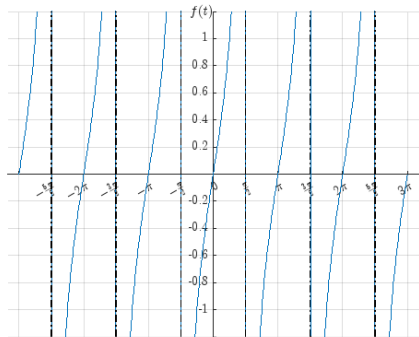
Sketching $\cos(t)$

$\cos(t) = 1$ when $t = 0$. It varies between -1 and 1 and takes 2π rad (or 360°) to complete one full cycle. It is identical to sine, but shifted left by $\pi/2$ radians.

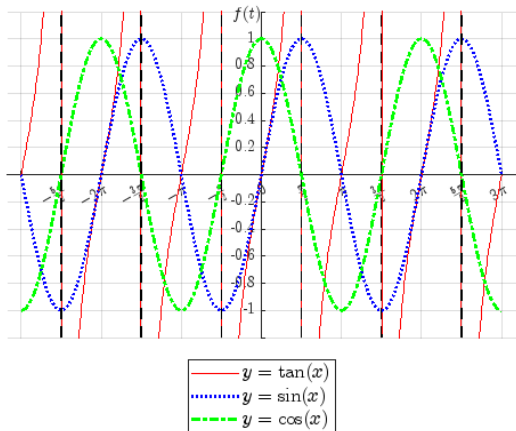


Sketching $\tan(t)$

$\tan(t) = 0$ when $t = 0$. It possesses asymptotes at $t = \pm\frac{\pi}{2}$ rad (or $\pm 90^\circ$) and takes π rad (or 180°) to complete one full cycle.

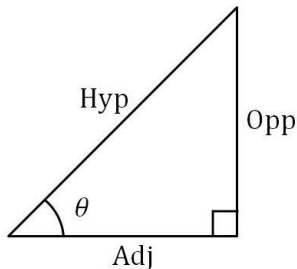


Sketching Sinusoids - Summary



Trigonometric Ratios: Right-angled Triangles

You may have encountered the trigonometric rules of a right-angled triangle:



Right-angled triangle trigonometric rules: **SOH-CAH-TOA**

$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

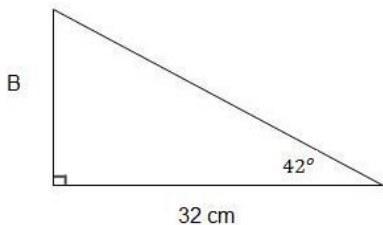
$$\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

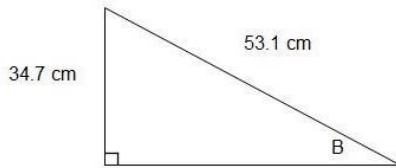
We can use these ratios to find unknown angles or side lengths.

Example 1

Calculate B in each of the following right-angled triangles:

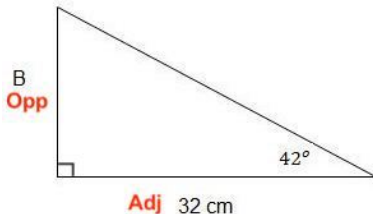


(a)



(b)

Example 1 - Solution (a)



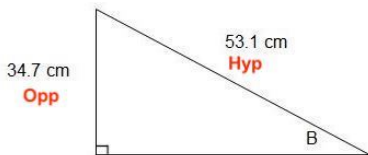
As the two sides of interest are the **o**pposite and the **a**djacent, we can use the **t**angent ratio (because of **TOA**):

$$\tan(42^\circ) = \frac{\text{Opp}}{\text{Adj}} = \frac{B}{32}$$

Transposing:

$$B = 32 \tan(42^\circ) = 28.81\text{cm} \quad (2 \text{ d.p.})$$

Example 1 - Solution (b)



As the two sides of interest are the **o**pposite and the **h**ypotenuse, we can use the **sine** ratio (because of **SOH**):

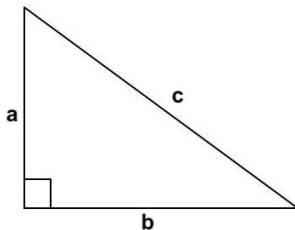
$$\sin(B^\circ) = \frac{\text{Opp}}{\text{Hyp}} = \frac{34.7}{53.1} = 0.65348 \dots$$

Using the inverse sine (or *arcsin*):

$$B = \sin^{-1}(0.65348) = 40.80^\circ \quad (2 \text{ d.p.})$$

Pythagoras' Theorem

In a right-angled triangle we can also calculate the length of a side, if the other two sides are known, using the **Pythagorean theorem**:



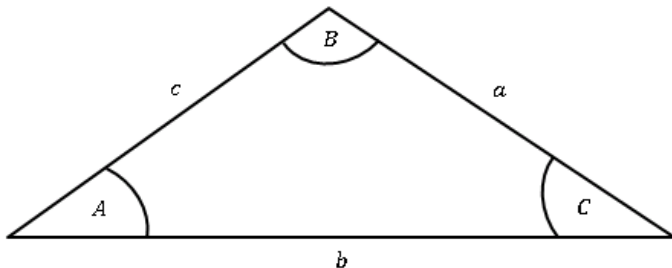
Pythagoras' Theorem

$$a^2 + b^2 = c^2$$

where c is the length of the hypotenuse.

Trigonometric Ratios: Non-right-angled Triangles

More general rules relate the sides and angles of non-right angled triangles:



There are two rules that can be used depending on what information we have and what requires calculation.

The Sine Rule

Sine Rule:

$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)} \quad (1)$$

Or, equivalently

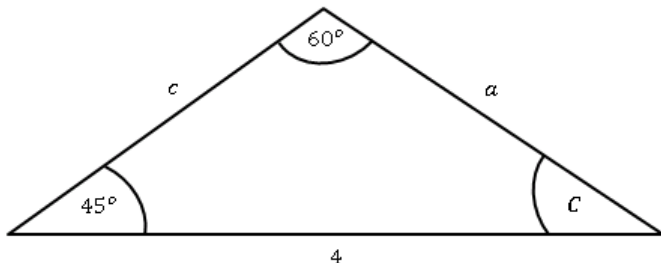
$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c} \quad (2)$$

These are equivalent, but it is easier to use equation (1) if a side is the unknown and equation (2) if an angle is the unknown.

Note: in order to use the sine rule, a complete pair must be known, i.e. a and A (a side and the angle facing it).

Example 2 (I/IV)

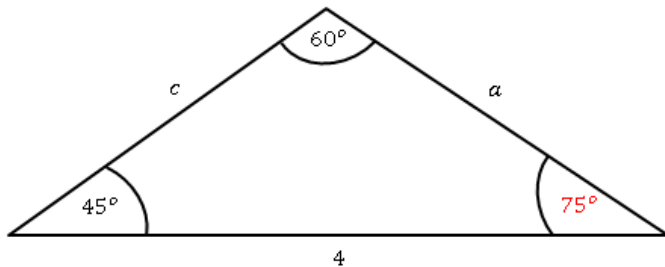
Find a , c and C .



First, calculate C by the fact that angles in a triangle sum to 180° .

$$\therefore C = 180 - 60 - 45 = 75^\circ.$$

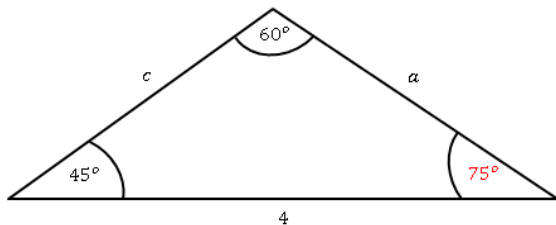
Example 2 (II/IV)



We have a complete pair (4 and 60°), so we can use the sine rule.

Let's use it to calculate a , as we also have the angle opposing it.

Example 2 (III/IV)



$$\frac{a}{\sin(A)} = \frac{b}{\sin(B)}$$

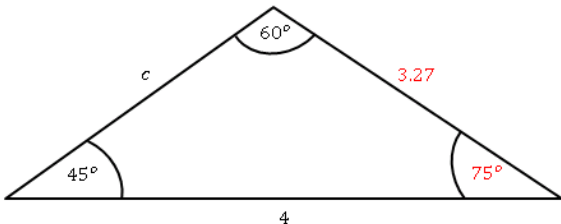
$$\frac{a}{\sin(45)} = \frac{4}{\sin(60)}$$

$$a = \sin(45) \times \frac{4}{\sin(60)}$$

$$\therefore a = 3.27 \text{ to 2 d.p.}$$

Example 2 (IV/IV)

Now for c :



$$\frac{b}{\sin(B)} = \frac{c}{\sin(C)}$$

$$\frac{4}{\sin(60)} = \frac{c}{\sin(75)}$$

$$c = \sin(75) \times \frac{4}{\sin(60)}$$

$$\therefore c = 4.46 \text{ to 2 d.p.}$$

The Cosine Rule

Cosine Rule:

$$a^2 = b^2 + c^2 - 2bc \cos(A) \quad (3)$$

Or

$$\cos(A) = \frac{b^2 + c^2 - a^2}{2bc} \quad (4)$$

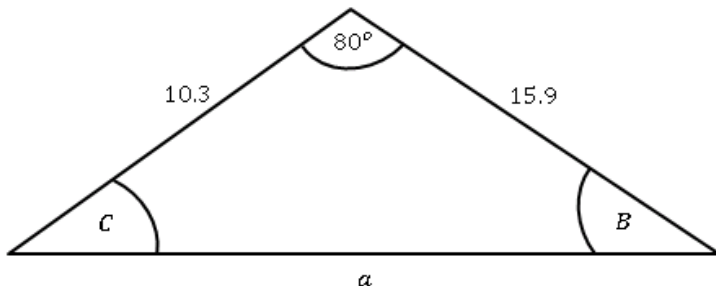
Equation (3) would be used to calculate a side and equation (4) would be used to calculate an angle.

Note: we use the cosine rule when we know either:

- **all three sides a, b, c or**
- **two sides b, c and the angle A inbetween them.**

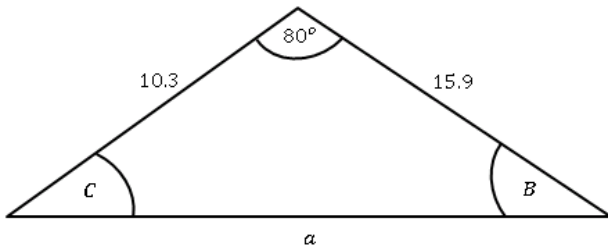
Example 3

Determine a :



Here, we don't have a complete pair so we cannot use the sine rule. However, we *do* know two sides and the angle (80°) inbetween, so we can use the cosine rule.

Example 3 - Solution



$$a^2 = b^2 + c^2 - 2bc \cos(A)$$

$$a^2 = 10.3^2 + 15.9^2 - 2 \times 10.3 \times 15.9 \times \cos(80)$$

$$a^2 = 302.0233$$

$$\therefore a = 17.38 \text{ to 2 d.p.}$$

The General Form

It is common in engineering to encounter quantities that vary in a sinusoidal fashion over time t (e.g. alternating currents). If y was such a quantity, we could say that:

General Form of a Sinusoidal Function:

$$y = A \sin(\omega t + \phi) + B$$

where A is the **amplitude**, B is the **vertical shift**, ω is the **angular frequency** and ϕ is the **phase shift**.

Amplitude A

Amplitude is the maximum distance from the midline (mean value) to a peak or trough.

Changing A has the visible effect of stretching the wave in the vertical axis.

- The standard sine wave $y = \sin(x)$ varies between -1 and +1, so the amplitude is exactly 1.
- $y = 2 \sin(x)$ varies between -2 and +2, so the amplitude is 2.
- As $y = \sin(x) + 3$ varies between 2 and 4, its amplitude is still just 1, as $\frac{4 - 2}{2} = 1$.

Vertical Shift B

The **vertical shift** (also known as the vertical offset or midline value) describes how much the wave is shifted up by, compared to the standard wave. We can find it by locating the vertical mean value of the graph.

- The standard sine wave $y = \sin(x)$ varies between -1 and +1, so the vertical shift is 0.
- $y = 2 \sin(x)$ varies between -2 and +2, so the vertical shift is still $\frac{-2 + 2}{2} = 0$.
- $y = \sin(x) + 3$ varies between 2 and 4, so in this case the vertical shift is $\frac{y_{\max} + y_{\min}}{2} = \frac{2 + 4}{2} = 3$.

Phase Shift ϕ

For $y = A \sin(\omega t + \phi) + B$ the **phase** constant ϕ shifts the graph horizontally. The actual horizontal displacement is

$$-\frac{\phi}{\omega}$$

For example, $y = \sin(x + 0.1)$ crosses the x -axis at $x = -0.1$, so the phase shift is 0.1 and the entire curve is shifted 0.1 to the left.

$y = \sin(x - 0.3)$ is shifted to the right by 0.3.

Angular Frequency ω

The **angular frequency** is the number of times the wave repeats in a distance of 2π along the horizontal axis. If the period (wavelength) is T the angular frequency is given by:

$$\omega = \frac{2\pi}{T}$$

The frequency is the number of complete wavelengths per unit along the horizontal axis: $f = T^{-1}$.

If the horizontal axis is time (seconds), then the frequency f units are Hertz (cycles/second) and the angular frequency is measured in radians/second.

Angular Frequency ω

Frequency and Period Formulae:

$$T = \frac{2\pi}{\omega} \quad T = \frac{1}{f} \quad \omega = 2\pi f$$

The angular frequency has the visual effect of compressing the wave along the horizontal axis, so that the new period is given by $T = \frac{2\pi}{\omega}$ rather than 2π .

We said earlier that the phase shift ϕ moves the graph horizontally. Combined with the effect of the angular frequency, together the graph moves by an amount $-\phi/\omega$.

Plotting with MATLAB

We can use MATLAB to easily plot a general trig. function with any values of the parameters A, ω, ϕ, B .

Here is an example of $f(t) = 2.3 \sin(0.2t - 1.3) + 4$:

```
syms f(t)  
f=2.3*sin(0.2*t-1.3)+4;  
fplot(f, 'k-', [-20 20])
```

Note that MATLAB expects angles in radians for the sin, cos and tan functions.

Solving Equations of The Form $A \sin(\omega x + \phi) + B = C$

We have seen how to solve equations involving many kinds of functions (quadratics, logarithms, etc.). How do we solve equations involving trigonometric functions?

Due to the periodicity of sine and cosine, an equation of the type $\sin(x) = c$ will either have no solutions (as sine only takes a certain range of values, i.e. $c \in [-1, 1]$) or infinitely-many solutions, unless we specify a restricted range of x that we are interested in.

For a general trigonometric equation $A \sin(\omega x + \phi) + B = C$, we follow a procedure to locate all solutions for x within a specified range.

Solving Equations of The Form $A \sin(\omega x + \phi) + B = C$

- 1 Define a new variable $u = \omega x + \phi$ to simplify the the trig. function to $\sin(u) = c$, where $c = (C - B)/A$.
- 2 Calculate the new range in terms of u , by substituting the limits of x into this formula.
- 3 Determine the set of solutions for u :
 - Use the inverse trigonometric function on your calculator to obtain the **principal value**:

$$u_0 = \sin^{-1}(c)$$

This is the first solution, and the value closest to the y -axis.

- For sine and cosine, use the symmetry of the graph to locate the **other** solution that occurs within the first cycle. This often takes the form $u_1 = \pi - u_0$ for sine, and $u_1 = -u_0$ for cosine.

Solving Equations of The Form $A \sin(\omega t + \phi) + B = C$

- To find all of the other solutions for u , then:
 - For sine and cosine, add and subtract integer multiples of 2π to **both** u_0 and u_1 until we are outside of the stated range.
 - For tangent, add and subtract multiples of π to u_0 .
 - Use high precision when calculating each solution, as errors may compound as we use u_0 to determine u_1 and then use that to determine subsequent solutions.

- ④ Convert solutions for u back to the corresponding solutions for x using:

$$x = \frac{u - \phi}{\omega}$$

- ⑤ Verify the final solutions by substituting back into $A \sin(\omega x + \phi) + B$ and evaluating.

Example

Solve

$$\sin(3x + 0.2) = 0.5$$

for $-\pi \leq x \leq \pi$.

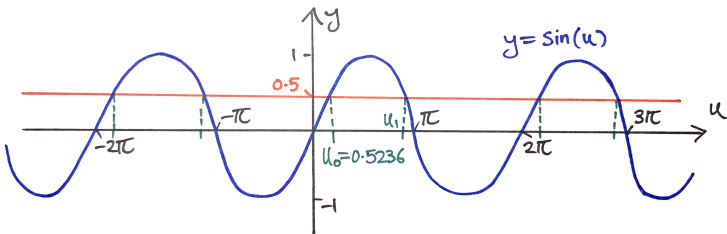
Solution (I/III)

Let $u = 3x + 0.2$, then the range is $-3\pi + 0.2 \leq u \leq 3\pi + 0.2$, or:

$$-9.2248 \leq u \leq 9.6248$$

Now the problem has been converted to:

“Solve $\sin(u) = 0.5$ for $-9.2248 \leq u \leq 9.6248$ ”.



Solution (II/III)

Obtain the principal value:

$$u_0 = \sin^{-1}(0.5) = \frac{\pi}{6} = 0.5236$$

From the symmetry of the graph, the other solution in the first period is:

$$u_1 = \pi - u_0 = \pi - 0.5236 = 2.6180$$

Adding and subtracting multiples of 2π , we find six solutions for u in the acceptable range. Then convert these back to solutions for x using:

$$x = \frac{u - 0.2}{3}$$

Solution (III/III)

u	In Range?	$x = \frac{u - 0.2}{3}$
$u_0 = 0.5236$	Yes	$(0.5236 - 0.2)/3 = 0.1079$
$u_0 + 2\pi = 0.5236 + 2\pi = 6.8068$	Yes	2.2022
$u_0 + 4\pi = 0.5236 + 4\pi = 13.090$	No	-
$u_0 - 2\pi = 0.5236 - 2\pi = -5.7596$	Yes	-1.9865
$u_0 - 4\pi = 0.5236 - 4\pi = -12.043$	No	-
$u_1 = 2.6180$	Yes	0.8060
$u_1 + 2\pi = 2.6180 + 2\pi = 8.9012$	Yes	2.9004
$u_1 - 2\pi = 2.6180 - 2\pi = -3.6652$	Yes	-1.2884
$u_1 - 4\pi = 2.6180 - 4\pi = -9.9484$	No	-

So we have six valid solutions:

$$x = -1.987, -1.288, 0.108, 0.806, 2.202, 2.900.$$